Md. Tanzamul Azad

AI Engineer | NLP & Generative AI Researcher
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Summary

Final-year CSE (Data Science) student researching trustworthy LLM systems (faithful reasoning, agent security, federated learning) and building production AI agents, RAG pipelines, and real-time platforms. Seeking research, graduate, teaching, and AI/ML roles.

Education

United International University (UIU), Dhaka, Bangladesh 2022 – Present
B.Sc. in Computer Science & Engineering, Major in Data Science | **CGPA: 3.78/4.00** | 50% Merit Scholarship
Core coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Linear Algebra, Statistics
HSC, Dhaka College, GPA 5.00/5.00 (2020) | **SSC**, Govt. Laboratory High School, GPA 5.00/5.00 (2018)

Research (Ongoing)

Interests: NLP, LLM reasoning & faithfulness, Generative AI, secure AI agents, federated learning, efficient edge AI.

- **Faithfulness-Checked Neuro-Symbolic Reasoning with LLMs** *In progress; target: ACL*
- **Effect-Level Security for MCP Tool Servers (LLM Agents)** *In preparation; target: USENIX Security 2027*
- **Energy-Efficient Federated Intrusion Detection for IoT** *In preparation; target: IEEE IoT Journal*
- **Camera-Free Indoor Scene Reconstruction (Embedded Sensing)** *In preparation; target: CVPR 2027*

Selected Projects

RadFlow: Offline AI Radiology Triage (Harvard HSIL Hackathon 2026) 2026
Full-Stack Developer | PyTorch, DenseNet121, GradCAM, CheXagent-8b, RAG (ChromaDB), FastAPI, React

- Built the FastAPI backend for a 4-stage chest X-ray pipeline (18-pathology detection, GradCAM localization, guideline retrieval, VLM report drafting) that runs fully offline on CUDA, Apple Silicon, or Intel hardware.
- Added an async inference queue with retries, latency/error monitoring, and a 7-day confidence-drift report.

Complete Legal Aid: AI Legal-Tech Platform (live) 2025 – 2026
Full-Stack Developer & AI Integration Lead | React, TypeScript, Django Channels, PostgreSQL, Gemini, FAISS, Docker

- Shipped a bilingual citizen–lawyer platform with four AI agents in production; the AI auto-drafting API cut document drafting from ∼45 to under 3 minutes (15×).
- Engineered the full frontend for 3 user roles and end-to-end WebSocket messaging, with JWT & Google OAuth 2.0.

ChefMate: NLP Recipe Retrieval (Solo) | Python, spaCy, scikit-learn, TF-IDF, kNN, Flask 2024

- Normalized noisy free-text ingredients with spaCy lemmatization and fuzzy matching, then ranked recipes from a 643 MB Kaggle dataset with TF-IDF + kNN and explainable match scores.

Care Companion: AI + IoT Health Rover (Team Lead) | Python, OpenCV, Raspberry Pi 4 2024

- Wrote all rover software: MobileNet-SSD person detection for autonomous patient tracking, vitals and gas sensing, voice commands, and a caregiver dashboard.

Also: ConvoVerse (multi-agent LLM HCI prototype) • AI Orchestrator (open-source contributor) • SkillEx (semantic matching, Spring Boot)

Teaching Experience

Academic Tutor (Independent) – Mathematics & Science, Classes 6–12; individual lesson plans Ongoing

Technical Skills

ML / AI: PyTorch, scikit-learn, spaCy, OpenCV, Pandas, NumPy, LLM APIs, RAG, FAISS, ChromaDB, AI Agents
Languages: Python, Java, C, C++, TypeScript, JavaScript, SQL | **Spoken:** Bangla (native), English (professional)
Engineering: FastAPI, Django, Flask, Spring Boot, React, PostgreSQL, Redis, Docker, GCP, Git, Linux, LaTeX

Honors & Awards

- **Champion**, UIU CSE Project Show (System Analysis & Design), Complete Legal Aid Summer 2025
- **3rd Runner-Up**, UIU CSE Project Show (Software Engineering), Complete Legal Aid Fall 2025
- **4th Runner-Up**, UIU CSE Project Show, Care Companion (AI + IoT) 2024
- **3rd Place**, National Math & Physics Olympiad, Junior Level (BdMOC) 2016

Activities

Delegate, Asian Youth MUN & AYIMUN (2018–19) • Member, British Council Bangladesh (2015–18) • Executive Member, School Quiz Club (2015–16)